Context-Aware Seq2Seq Prompt Prediction

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Problem and Motivation

Our project works backwards: Given a ChatGPT response, predict what the original user prompt was. We created a Sequence-to-Sequence NLP model where the input is an AI-generated answer, and the output is the predicted prompt.

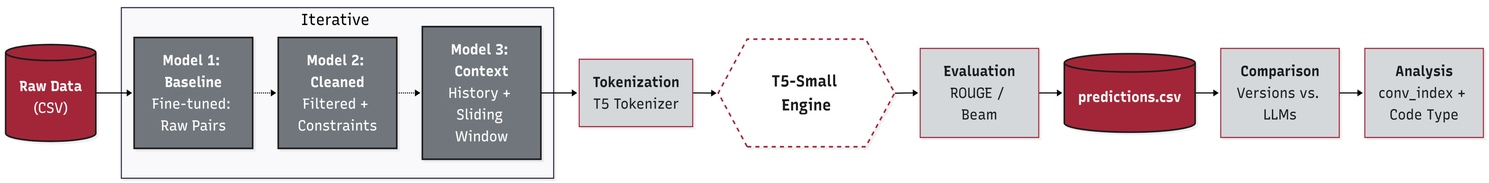
Prompt Steering is essential to many larger topics in AI and LLM. Gain an understanding of how to structure prompts to obtain the best answer.

By truly understanding how to manipulate LLMs through reverse-engineering prompt steering, our capabilities will increase drastically.

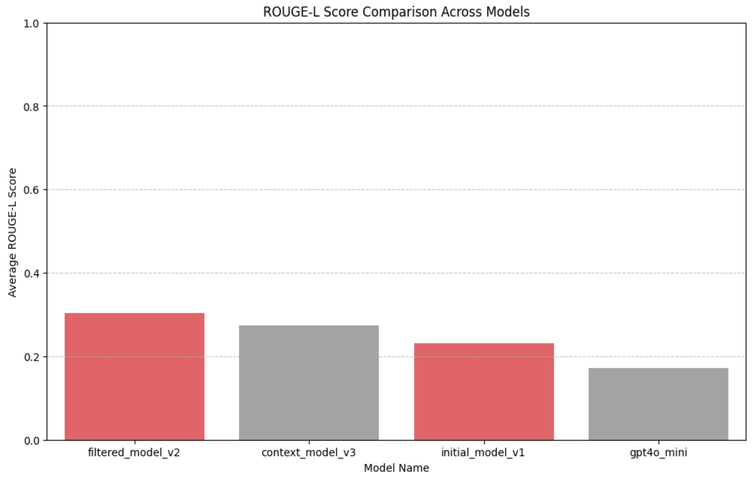
Field of Study

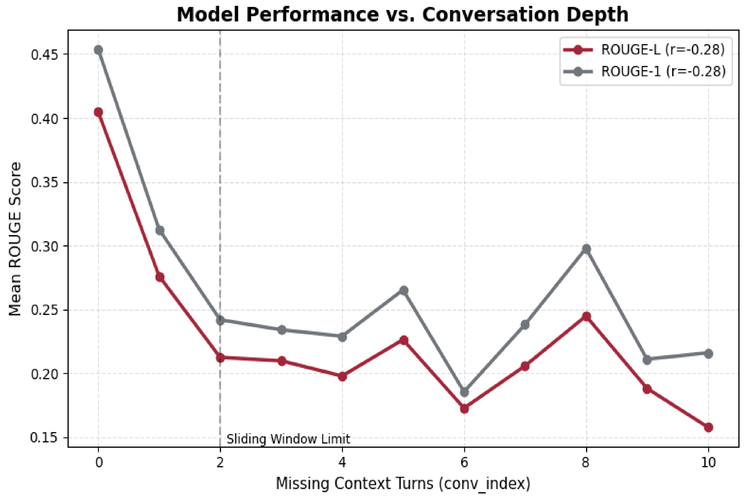
	Du et al. (2017) [1]	Raffel et al. (2020) [2]	Our Project
Architecture	Recurrent Seq2Seq	Unified transformer	Fine-tuned T5 model
Data Handling	Sentence vs. paragraph encoding trade-offs	Pre-trained on C4 dataset, task-specific prefixing	Filtered out programming noise, 2-turn history context window
Training & Optimization	Maximized likelihood of target question given source text using SGD	Denoisng objective, supervised fine-tuning	AdamW, gradient clipping and early stopping for stability
Key Findings	Neural models beat rule-based complexity and naturalness	Increasing model/data size leads to great results	ROUGE scores inversely correlate with dialogue depth

Method/Approach



Experiments and Results





	Low Performance (ROUGE-L: 0.25)	Medium Performance (ROUGE-L: 0.51)	High Performance (ROUGE-L: 0.90)
Turn	8 (Deep Context)	4 (Mid Context)	1 (Start)
Actual Prompt	"Can you do the same for CAVA?"	"How do I change my primary monitor on common Oses?"	"What is EDI 945?
Model Prediction	"Summarize news about Sweetgreen"	"How do I set a primary display on the three most common operating systems?"	"What is the EDI 945?"

Analysis/Discussion

When adding 1-2 messages of previous context, our model performed slightly worse. This may be because the prompt and answers became very long. It was difficult for the model to recognize which parts were truly relevant to only the most recent prompt.

As the conversation progresses, the model's accuracy drops. Once the conversation exceeds the two-turn history, the T5 model loses global context. The prefix becomes noisier as the conversation deepens.

When the model gets confused, it defaults to generic prompts. This indicates the T5 model lacks the reasoning depth to reconstruct complex intent without a larger history.

Conclusion and Future Work

Overall, our models do quite an excellent job of predicting prompts, even beating ChatGPT's 4.0 mini by a considerable amount.

Implementing the model was fundamentally time-consuming since our model was continuously overfitting. This caused us to originally think it was performing well, while it was really performing poorly. We later discovered that this was due to too much context, which created a large grey area for us. In the future, we will experiment with only keeping the keywords (or a small summary) from the prior context to prevent input length from growing too large.

Another major addition we can add to this project in the future would be changing to a larger scale model, like T5-Base/Llama variants.

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References

[1] Neural Question Generation [Du et al., ACL 2017].

[2] Limits of Transfer Learning [Raffel et al., JMLR 2020]

[3] Context-Aware Generation [Dong et al., U2BigData 2024]

[4] Prompt Engineering for LLMs [Medium, 2026]

[5] Sliding Window Attention [Fu et al. 2025]

[6] Steering Target Atoms for LLM Control [Wang et al., ACL 2025]

[7] Prompt Steerability of LLMs [Miehling et al., NAACL 2025]